

Part A · Multiple Choice

2 points each

1. **a) The number of protons in the nucleus**

p. 116 — all atoms of a particular element are identical, and the proton count is what fixes the identity.

2. **d) a cation with a positive charge**

p. 117 — losing a negatively charged electron leaves more protons than electrons, so the net charge is positive. Gaining an electron produces an anion.

3. **c) Silicates**

p. 116 — carbonates and native elements are also named as important, but chiefly for their usefulness to society rather than their abundance.

4. **a) streak does not vary**

p. 119 — two minerals of the same color may leave different colored streaks. Pyrite looks gold but streaks blackish, while real gold streaks golden yellow.

5. **b) Talc**

p. 121 — talc can be scratched by all other materials. The full scale runs talc, gypsum, calcite, fluorite, apatite, feldspar, quartz, topaz, corundum, diamond.

6. **b) cleavage**

pp. 121–122 — cleavage follows lines of weakness in the mineral's structure. Calcite cleaves in three directions into rhombs, and galena into true cubes.

Part B · True or False

1 point each

7. **FALSE**

Color is the most eye-catching property but usually the least reliable. Pure quartz is colorless, while amethyst is the same mineral colored purple by traces of iron. (pp. 117–118)

8. **TRUE**

p. 123 — unlike cleavage, which follows structural planes of weakness. Quartz is known for fracturing into smooth curved surfaces.

Part C · Short Answer

3 points each

9. **It must occur naturally, be solid, have an ordered internal (crystallographic) structure, and have a fairly well-defined chemical composition.**

Rubric: 1 pt for each of any three. (1) Naturally occurring, formed by natural geological processes — this excludes anything made by human activity or by living beings, such as seashells or tungsten carbide. (2) A solid in its natural occurrence. (3) A well-defined crystallographic structure, or more generally an ordered atomic arrangement. (4) A fairly well-defined chemical composition. (p. 124)

10. Cleavage is the tendency of a mineral to break along structural planes of weakness, producing smooth surfaces. Fracture is any break that does not follow a cleavage plane.

Rubric: 1 pt cleavage defined — breakage along certain planes determined by the mineral's internal structure, yielding smooth surfaces. 1 pt fracture defined — a break *not* along a cleavage plane, and therefore not fixed by structure. 1 pt for one valid example of each. Cleavage: mica (one plane, thin sheets), calcite (three planes, rhombs), galena or halite (three planes at right angles, cubes). Fracture: quartz (smooth curved surfaces), chrysotile (splintery), metals (jagged edges). (pp. 121–123)